

Translated

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Sqrt Decomposition

Sqrt Decomposition is a method (or a data structure) that allows you to perform some common operations (finding sum of the elements of the sub-array, finding the minimal/maximal element, etc.) in $O(\sqrt{n})$ operations, which is much faster than $O(n)$ for the trivial algorithm.

First we describe the data structure for one of the simplest applications of this idea, then show how to generalize it to solve some other problems, and finally look at a slightly different use of this idea: splitting the input requests into sqrt blocks.

Sqrt-decomposition based data structure

Given an array $a[0 \dots n-1]$, implement a data structure that allows to find the sum of the elements $a[l \dots r]$ for arbitrary l and r in $O(\sqrt{n})$ operations.

Description

The basic idea of sqrt decomposition is preprocessing. We'll divide the array a into blocks of length approximately $\sqrt{n}$, and for each block i we'll precalculate the sum of elements in it $b[i]$.

We can assume that both the size of the block and the number of blocks are equal to $\sqrt{n}$ rounded up:

$$s = \lceil \sqrt{n} \rceil$$

Then the array a is divided into blocks in the following way:

$$\underbrace{a[0], a[1], \dots, a[s-1]}_{\text{b[0]}}, \underbrace{a[s], \dots, a[2s-1]}_{\text{b[1]}}, \dots, \underbrace{a[(s-1) \cdot s], \dots, a[n-1]}_{\text{b[s-1]}}$$

The last block may have fewer elements than the others (if n not a multiple of s), it is not important to the discussion (as it can be handled easily). Thus, for each block k , we know the sum of elements on it $b[k]$:

$$b[k] = \sum_{i=k \cdot s}^{\min\{(n-1), (k+1) \cdot s - 1\}} a[i]$$